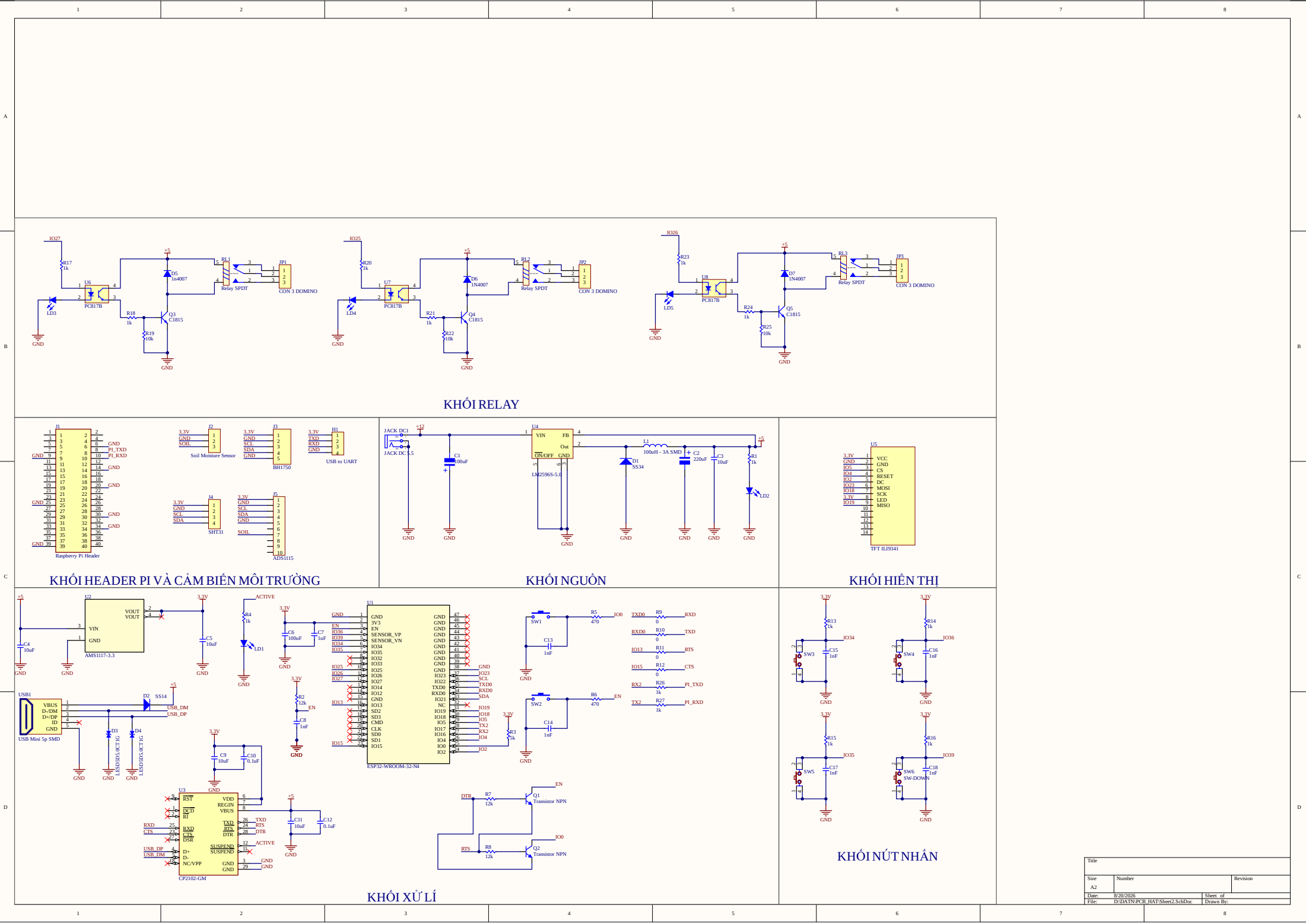




KHỎI RELAY

KHỎI NGUỒN

KHỎI HIỂN THỊ

KHỎI HEADER PI VÀ CẢM BIẾN MÔI TRƯỜNG

KHỎI NÚT NHẤN

KHỎI XỬ LÝ

|       |                               |           |
|-------|-------------------------------|-----------|
| Title |                               |           |
| Size  | Number                        | Revision  |
| A2    |                               |           |
| Date: | 8/20/2026                     | Sheet of  |
| File: | D:\DATN\PCB_HAT\Sheet2_SchDoc | Drawn By: |

| Comment                     | Description                                                   | Designator                                                         | Footprint                   | LibRef                      | Quantity |
|-----------------------------|---------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------|-----------------------------|----------|
| 100uF                       | Dán                                                           | C1                                                                 | tu nhóm 6.3 (d)             | Tu Nhóm 100uF 16V           | 1        |
| 220uF                       | Dán                                                           | C2                                                                 | tu nhóm 8 (e)               | Tu Nhóm 220uF 10V           | 1        |
| 10uF                        | Capacitor SMD 0603                                            | C3, C4, C5, C9, C11                                                | Cap 0603                    | Cap 0603                    | 5        |
| 100uF                       | Capacitor SMD 0603                                            | C6                                                                 | Cap 0603                    | Cap 0603                    | 1        |
| 1uF                         | Capacitor SMD 0603                                            | C7                                                                 | Cap 0603                    | Cap 0603                    | 1        |
| 1nF                         | Capacitor SMD 0603                                            | C8, C13, C14, C15, C16, C17, C18                                   | Cap 0603                    | Cap 0603                    | 7        |
| 0.1uF                       | Capacitor SMD 0603                                            | C10, C12                                                           | Cap 0603                    | Cap 0603                    | 2        |
| SS34                        | Dán                                                           | D1                                                                 | DIODE SMA                   | Diode SS34 ( 1N5822 )       | 1        |
| SS14                        | Dán                                                           | D2                                                                 | DIODE SMA                   | Diode SS14 ( 1N5819)        | 1        |
| LES5D5.0CT1G                | Transient Voltage Suppressors for ESD Protection              | D3, D4                                                             | TVS_LES5D5.0CT1G            | LES5D5.0CT1G                | 2        |
| 1n4007                      | Cắm                                                           | D5, D6, D7                                                         | DIODE 1A                    | Diode 1N4007                | 3        |
| USB to UART                 | Header IDC, 4 x 1 Pin Vertical 2.54mm Pitch (Non-Boxed, Male) | H1                                                                 | HDR4                        | Header4                     | 1        |
| Raspberry Pi Header         | 40 Position Header Connector 0.100 (2.54mm) Through Hole Gold | J1                                                                 | SAMESKY_SSK04-FPH-254-B20   | SSK04-FPH-254-B20           | 1        |
| Soil Moisture Sensor        | WR-PHD 2.54 mm Angled Socket Header                           | J2                                                                 | JP3-C1T                     | JP3                         | 1        |
| BH1750                      | WR-PHD 2.54 mm Angled Socket Header                           | J3                                                                 | JP5-C1T                     | JP5                         | 1        |
| SHT31                       | WR-PHD 2.54 mm Angled Socket Header                           | J4                                                                 | JP4-C1T                     | JP4                         | 1        |
| ADS1115                     | Connector 10                                                  | J5                                                                 | CON10                       | Con10                       | 1        |
| JACK DC 5.5                 | JACK DC 5.5                                                   | JACK DC1                                                           | JACK DC 5.5                 | JACK DC 5.5                 | 1        |
| CON 3 DOMINO                |                                                               | JP1, JP2, JP3                                                      | CON 3                       | CON 3 DOMINO                | 3        |
| 100uH - 3A SMD              | Dán                                                           | L1                                                                 | Cuon Cam 12x12x10           | CUON CAM 100uH SMD          | 1        |
| Led                         | Typical LED                                                   | LD1, LD2, LD3, LD4, LD5                                            | Led 0805                    | Led                         | 5        |
| Transistor NPN              | NPN Bipolar Transistor                                        | Q1, Q2                                                             | SOT-23                      | Transistor NPN              | 2        |
| C1815                       | NPN 600mW, ECB                                                | Q3, Q4, Q5                                                         | TO92-123                    | 2N222                       | 3        |
| 1k                          | RES SMD 0603                                                  | R1, R4, R13, R14, R15, R16, R17, R18, R20, R21, R23, R24, R26, R27 | RES 0603                    | RES 0603                    | 14       |
| 12k                         | RES SMD 0603                                                  | R2, R7, R8                                                         | RES 0603                    | RES 0603                    | 3        |
| 5k                          | RES SMD 0603                                                  | R3                                                                 | RES 0603                    | RES 0603                    | 1        |
| 470                         | RES SMD 0603                                                  | R5, R6                                                             | RES 0603                    | RES 0603                    | 2        |
| 0                           | RES SMD 0603                                                  | R9, R10, R11, R12                                                  | RES 0603                    | RES 0603                    | 4        |
| 10k                         | RES SMD 0603                                                  | R19, R22, R25                                                      | RES 0603                    | RES 0603                    | 3        |
| Relay SPDT                  | Relay SPDT                                                    | RL1, RL2, RL3                                                      | RL_SPDT_10A                 | Relay SPDT                  | 3        |
| Nut Nhan 2 Chan SMD 3x6x2.5 | Nut Nhan 2 Chan SMD 3x6x2.5                                   | SW1, SW2                                                           | Nut Nhan 2 Chan SMD 3x6x2.5 | Nut Nhan 2 Chan SMD 3x6x2.5 | 2        |
| SW_CAPTURE                  | Switch button 4 pin                                           | SW3                                                                | SW2-DIP                     | SW4                         | 1        |
| SW_UP                       | Switch button 4 pin                                           | SW4                                                                | SW2-DIP                     | SW4                         | 1        |
| SW-OK                       | Switch button 4 pin                                           | SW5                                                                | SW2-DIP                     | SW4                         | 1        |
| SW-DOWN                     | Switch button 4 pin                                           | SW6                                                                | SW2-DIP                     | SW4                         | 1        |
| ESP32-WROOM-32-N4           | Description                                                   | U1                                                                 | MOD_ESP32-WROOM-32_EXP      | ESP32-WROOM-32-N4           | 1        |
| AMS1117-3.3                 | Description                                                   | U2                                                                 | SOT-223 UMW                 | AMS1117-3.3                 | 1        |
| CP2102-GM                   | USB Bridge, USB to UART USB 2.0 UART Interface 28-QFN (5x5)   | U3                                                                 | QFN50P500X500X100-29N       | CP2102-GM                   | 1        |
| LM2596S-5.0                 | Simple Switcher 3A Step-Down Voltage Regulator                | U4                                                                 | TO-263-5                    | LM2596S-5.0                 | 1        |
| TFT ILI9341                 |                                                               | U5                                                                 | TFT ILI9341                 | TFT ILI9341                 | 1        |
| PC817B                      |                                                               | U6, U7, U8                                                         | DIP 4                       | Opto PC817B                 | 3        |
| USB Mini 5p SMD             | USB Type B                                                    | USB1                                                               | USB Mini 5p SMD             | USB Mini 5p SMD             | 1        |